few remedial measures. After the eye-opener at the LoC, Major General C.S. Mann, additional director general with the Army Design Bureau (ADB), has been framing a methodology to block the use of Chinese equipment in the military and has been meeting all stakeholders, including UAV manufacturers. Once finalised, the framework will be submitted to the MoD. The finalised drone procurement framework will be applied to the Indian Air Force, Navy and Coast Guard too.

Maj. Gen. Mann says that any electronic part has the potential for data transfer. "With supply chains heavily reliant on Chinese components, the focus is on finding solutions to eliminate their presence. Various strategies are under deliberation, and a comprehensive framework will soon be developed," he says.

THREATS AND LOOPHOLES

Experts classify Chinese components in drones into two types—active and passive. Active components, which include processors, communication modules, sensors, flight control units and memory storage devices are critical to drone operations and highly vulnerable to cyber threats. For example, processors with malicious firmware (microcode or program) can help adversaries intercept surveillance data or manipulate drone functions; communication modules and sensors can transmit sensitive reconnaissance data to foreign servers. Passive components like printed circuit boards (PCBs), motors and power systems also carry risks. PCBs can conceal spyware or sabotage mechanisms, compromised motors and power systems can lead to premature failures and substandard resistors or capacitors increase system failure risks during missions.

Defence analysts identify specific threats too. Injection of malware, where components are embedded with malicious programs during manufacturing, disrupting operations and compromising data security, is one. Components like cameras, LiDAR (Light Detection and Ranging) sensors and communication modules can also be rigged with malware to transmit information to adversaries. Furthermore, a dependency on Chinese systems limits India's ability to customise drones for defence needs. Given its domination in drone components, Beijing can manipulate their supply chain to disrupt Indian drone production and operational readiness.

"The primary reason for the hijacking of Indian assembled drones is the use of vulnerable open source material whose survivability is poor because it was never intended to be used for commercial and security applications," says Sai Pattabiram, founder and MD of Chennai-based ZUPPA Geo Navigation Technologies Pvt. Ltd. His company works closely with the Indian Army to provide it with hacking-proof drones. He adds that the rise of drone swarms, loitering munitions (armed drones that 'loiter' in the sky before locating the target) and autonomous UCAVs means that defence forces must adapt to the growing cyber threats that accompany these technologies. He also points out that the rampant use of COTS (commercial off-the-shelf) civilian drones in recent conflicts means that both civilian and defence drones require similar levels of cybersecurity measures.

Addressing the vulnerabilities posed by Chinese components in drones will be crucial to maintaining operational security and national sovereignty, says another MoD official. "The government is expected to expedite programmes aimed at indigenous development of UAVs," he adds, claiming that a pilot project is going to be launched soon to check how supply chain issues can be addressed.

Group Captain R.K. Narang, a drone expert who has worked extensively with Indian drone manufacturers, blames flawed government policy. He says that the Defence Acquisition Procedure-2020 (DAP-2020) mandates that Indigenous Designed, Developed, and Manufactured (IDDM) products with more than 50 per cent Indigenous Content (IC) be given preference in procurement to promote Indian firms. However, due to loopholes, assembled products with negligible indigenous content could qualify as IDDM. A few manufacturers also get cheap Chinese equipment allegedly by routing them through their subsidiary firms based

FOREIGN COMPONENTS IN INDIAN DRONES COULD ENABLE FUNCTION MANIPULATION, EXTRACTION OF SENSITIVE MILITARY INFORMATION, OR SYSTEM FAILURE DURING MISSIONS

DANGER IN THE SKIES

Chinese components used in some Indian military drones that could potentially be compromised



NAVIGATION SYSTEMS

Systems, including GPS and inertial navigation systems, can be corrupted by, say, false coordinates.

Firms: Beijing BDStar Navigation, Huace Navigation

SENSORS AND CAMERAS

Some Indian drones use Chinese sensors and cameras that are ripe for illegal transmission of data through malicious firmware

Firms: Hangzhou Hikvision, Zhejiang Dahua

